SHIBAJI CHAKRABORTY

Ph.D. | Research Scientist & Data Scientist

Space Weather · Ionospheric Physics · HF Propagation · ML/AI

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EDUCATION

Institution / Degree	Period
Virginia Polytechnic Institute and State University (Virginia Tech) Ph.D. in Electrical Engineering (thesis)	*08/2015 – 05/2021*
Virginia Polytechnic Institute and State University (Virginia Tech) M.A. in Data Analytics and Applied Statistics	*08/2019 – 12/2020*
Virginia Polytechnic Institute and State University (Virginia Tech) M.S. in Electrical Engineering	*08/2015 – 05/2017*
West Bengal University of Technology, Kolkata, India (WBUT) B.Tech. in Electronics and Communication Engineering	*08/2006 – 05/2010*

APPOINTMENTS

Institution	Role	Period
Embry-Riddle Aeronautical University	**Research Scientist**	*09/2024 – Present*
Virginia Tech	**Postdoctoral Associate**	*05/2021 – 09/2024*
Tata Consultancy Services	**Information Technology Analyst**	*09/2010 – 07/2015*

RESEARCH GRANTS

Grant / Project Title	Period
Co-I (Institutional PI) of NASA LWS Research Award – 'Ionospheric responses to thunderstorm-generated acoustic and gravity waves over the continental US'	*2022 – 2026*
PI of NSF Award – 'Collaborative Proposal: Investigation of ionospheric density response to American Solar Eclipses using coordinated radio observations with modeling support'	*2024 – 2027*
Co-I of NASA Research Award – 'Determining the connections between driving conditions, magnetosphere-ionosphere-ground properties, and extreme geomagnetic and geoelectric disturbances'	*2024 – 2025*
PI of NSF-GEM Research Award – 'Collaborative Research: GEM: Modeling Ionospheric and Magnetospheric Current Interactions with Submarine Cables'	*2024 – 2027*

FELLOWSHIPS & AWARDS

Institution	Award / Fellowship	Year
Google	Google Academic Research Award	*2025*
JSPS (Japan)	JSPS Postdoctoral Fellowship (Short-term)	*2022*
Nagoya University	ISEE International Joint Research Fellowship	*2022*
IEEE/WiSEE	Best Conference Paper	*2021*

NCAR/UCAR	ASP-GVP Fellowship	*2020*
Virginia Tech	Pratt Fellowship	*2019*
LANL/ISR	Vella Fellowship and ISR-1 Summer School	*2018*
Techno India	Won 'Electronically Yours' event in category 'Newron'	*2010*

PROFESSIONAL SERVICES

Organization	Role / Activity	Since / Year
AGU Fall Meeting	Primary Convener	*2022*
AGU Fall Meeting	OSPA Judge	*Since 2021*
NASA	Panel Reviewer of NASA Program	*2023*
Virginia Tech	Mentor of Google Summer of Code	*2018*
SuperDARN	Active member of SuperDARN Data Visualization Working Group	*Since 2021*
Reviewer	Nature Communication; JGR: Space Physics; Space Weather; Radio Science; Astrophysical Journal; Earth Planet and Science; Transactions on Geoscience and Remote Sensing; Advances in Space Research; EGU Sphere; Journal of Space Weather and Space Climate	*Since 2019*
NSF	Panel Reviewer of NSF Program	*2024*
CEDAR	Judge	*2022 / 2024*

JOURNAL PUBLICATIONS

1. Chakraborty, S., Hartinger, M. D., Shi, X., Boteler, D. H., Lawrence, E., Boteler, C., et al. (2026). Validating SCUBAS predictions of geomagnetically induced voltage in submarine cables using legacy superstorm observations. Space Weather, 24, e2025SW004843. https://doi.org/10.1029/2025SW004843
2. Bergsson, B., Inchin, P., Debchoudhury, S., Chakraborty, S., Snively, J., & Zettergren, M. (2026). Automated geolocation of diverse impulsive events using total electron content measurements. *Frontiers in Astronomy and Space Sciences*, *13–2026*. doi:10.3389/fspas.2026.1812901
3. Chakraborty, S., Bullett, T., Barjatya, A., & Mabie, J. (2026). pynasonde: An open-source Python library for ionosonde data processing. *SoftwareX*, *34*, 102617. https://doi.org/10.1016/j.softx.2026.102617
4. Wu, H., Wang, W., Shi, X., Lin, D., Hartinger, M. D., Baker, J. B. H., & Chakraborty, S. (2026). Global impacts of ultra-low-frequency waves: 1. Thermospheric responses and traveling atmospheric disturbances. Geophysical Research Letters, 53, e2025GL119835. https://doi.org/10.1029/2025GL119835
5. Chakraborty, S., Nishitani, N., Shi, X., Ponomarenko, P., Ruohoniemi, J. M., Baker, J. B. H., Coster, A. J., & Häggström, I. (2025). Solar Flare-Induced Gradient Drift Instability Observed by SuperDARN HF Radars. Journal of Geophysical Research: Space Physics, 130(10), e2025JA033824.
6. Bergsson, B., Debchoudhury, S., Inchin, P., Heale, C., Chakraborty, S., & Zettergren, M. (2025). The Impact of Receiver-Satellite Line-of-Sight Orientation on GNSS TEC Observations of TIDs Induced by Gravity Waves From Convective Sources. Journal of Geophysical Research: Space Physics, 130(9), e2025JA033938.
7. Inchin, P. A., Heale, C., Zettergren, M. D., Bergsson, B., Debchoudhury, S., & Chakraborty, S. (2025). Severe weather-generated acoustic and gravity wave impacts on the ionosphere: A model-guided case study. Journal of Geophysical Research: Space Physics, 130, e2025JA034012. https://doi.org/10.1029/2025JA034012
8. Prikryl, P., Themens, D. R., Chum, J., Chakraborty, S., Gillies, R. G., and Weygand, J. M. (2025). Observations of traveling ionospheric disturbances driven by gravity waves from sources in the upper and lower atmosphere. Ann. Geophys., 43, 511-534, https://doi.org/10.5194/angeo-43-511-2025
9. Chakraborty, S., Qian, L., Mrak, S., Mabie, J., Goncharenko, L., McInerney, J. M., & Bullett, T. (2025). Formation of the ionospheric G-condition following the 2017 Great American Eclipse. Earth and Space Science, 12, e2024EA004007. https://doi.org/10.1029/2024EA004007
10. Oberheide, J., Aggarwal, D., Bergsson, B., Chakraborty, S. et al. (2025). Impact of Terrestrial Weather on the Space Weather of the Ionosphere-Thermosphere: Initial Results from a NASA Living with a Star Focused Science Topic. Surv Geophys. https://doi.org/10.1007/s10712-025-09895-7
11. Rout, D., Kumar, A., Singh, R., Patra, S., Karan, D. K., Chakraborty, S., et al. (2025). Evidence of unusually strong equatorial ionization anomaly at three local time sectors during the Mother's Day geomagnetic storm on 10-11 May 2024. Geophysical Research Letters, 52, e2024GL111269. https://doi.org/10.1029/2024GL111269

12. Shi, X., Chakraborty, S., Baker, J. B. H., Hartinger, M. D., Wang, W., Ruohoniemi, J. M., et al. (2025). Statistical characterization of joule heating associated with ionospheric ULF perturbations using SuperDARN data. Journal of Geophysical Research: Space Physics, 130, e2024JA033452. https://doi.org/10.1029/2024JA033452
13. Liu, X., Zhang, D., Coster, A. J., Xu, Z., Shi, X., & Chakraborty, S. (2024). The morphology and oscillations of nightside mid-latitude ionospheric trough at designated longitudes in the Northern Hemisphere. Journal of Geophysical Research: Space Physics, 129, e2024JA032864. https://doi.org/10.1029/2024JA032864
14. Inchin, P. A., Bhatt, A., Bramberger, M., Chakraborty, S., Debchoudhury, S., & Heale, C. (2024). Atmospheric and ionospheric responses to orographic gravity waves prior to the December 2022 cold air outbreak. Journal of Geophysical Research: Space Physics, 129, e2024JA032485. https://doi.org/10.1029/2024JA032485
15. Gowtam, V. S., Connor, H., Kunduri, B. S. R., Raeder, J., Laundal, K. M., Tulasi Ram, S., Chakraborty S., et al. (2024). Calculating the high-latitude ionospheric electrodynamics using a machine learning-based field-aligned current model. Space Weather, 22, e2023SW003683. https://doi.org/10.1029/2023SW003683
16. Boteler, D. H., Chakraborty, S., Shi, X., Hartinger, M. D., & Wang, X. (2024). An examination of geomagnetic induction in submarine cables. Space Weather, 22, e2023SW003687. https://doi.org/10.1029/2023SW003687
17. Coyle, S. E., Baker, J. B. H., Chakraborty, S., Hartinger, M. D., Freeman, M. P., Clauer, C. R., et al. (2023). Substorms and solar eclipses: A mutual information-based study. Geophysical Research Letters, 50, e2023GL106432. https://doi.org/10.1029/2023GL106432
18. Reiss, M. A., Muglach, K., Mason, E., Davies, E. E., Chakraborty, S., Delouille, V., ... & Veronig, A. (2023). A Community Dataset for Comparing Automated Coronal Hole Detection Schemes. Astrophysical Journal Supplement. https://ore.exeter.ac.uk/repository/handle/10871/134745
19. Boteler, D., Chakraborty, S., Shi, X., Hartinger, M. and Wang, X. (2023). Transmission Line Modelling of Geomagnetic Induction in the Ocean/Earth Conductivity Structure. International Journal of Geosciences, 14, 767-791. doi: 10.4236/ijg.2023.148041
20. Rout, D., Patra, S., Kumar, S., Chakrabarty, D., Reeves, G. D., Stolle, C., Pandey, K., Chakraborty, S., Spencer E. A. (2023). The growth of ring current/SYM-H under northward IMF Bz conditions present during the 21-22 January 2005 geomagnetic storm. Space Weather, 21, e2023SW003489. https://doi.org/10.1029/2023SW003489
21. Liu, J., Chakraborty, S., Chen, X., Wang, Z., He, F., Hu, Z., et al. (2023). Transient response of polar-cusp ionosphere to an interplanetary shock. Journal of Geophysical Research: Space Physics, 128, e2022JA030565. doi: 10.1029/2022JA030565
22. Collins, K., Gibbons, J., Frissell, N., Montare, A., Kazdan, D., Kalmbach, D., Swartz, D., Benedict, R., Romanek, V., Boedicker, R., Liles, W., Engelke, W., McGaw, D. G., Farmer, J., Mikitin, G., Hobart, J., Kavanagh, G., and Chakraborty, S. (2023). Crowdsourced Doppler measurements of time standard stations demonstrating ionospheric variability. Earth Syst. Sci. Data, 15, 1403-1418, doi:10.5194/essd-15-1403-2023
23. Chakraborty, S., Boteler DH, Shi X, Murphy BS, Hartinger MD, Wang X, Lucas G and Baker JBH (2022). Modeling geomagnetic induction in submarine cables. Front. Phys. 10:1022475. doi: 10.3389/fphy.2022.1022475
24. Prikryl, P., Gillies, R. G., Themens, D. R., Weygand, J. M., Thomas, E. G., and Chakraborty, S. (2022). Multi-instrument observations of polar cap patches and traveling ionospheric disturbances generated by solar wind Alfven waves coupling to the dayside magnetosphere. Ann. Geophys., 40, 619-639, doi: 10.5194/angeo-40-619-2022
25. Chakraborty, S., Qian, L., Baker, J. B. H., Ruohoniemi, J. M., Kuyeng, K., and McInerney, J. M. (2022). Driving influences of the Doppler flash observed by SuperDARN HF radars in response to solar flares. Journal of Geophysical Research: Space Physics, 127, e2022JA030342. doi: 10.1029/2022JA030342
26. Shi X, Schmidt M, Martin CJ, Billett DD, Bland E, Tholley FH, Frissell NA, Khanal K, Coyle S, Chakraborty S, Detwiller M, Kunduri B and McWilliams K (2022). pyDARN: A Python software for visualizing SuperDARN radar data. Front. Astron. Space Sci. 9:1022690. doi: 10.3389/fspas.2022.1022690
27. Chakraborty, S., Baker, J. B. H., and Ruohoniemi, J. M. (2021). Probabilistic Short-wave Fadeout Detection in SuperDARN Time Series Observations, 2021 IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE), pp. 43-48, doi: 10.1109/WiSEE50203.2021.9613835
28. Chakraborty, S., Qian, L., Ruohoniemi, J. M., Baker, J. B. H., McInerney, J. M., and Nishitani, N. (2021). The role of flare-driven ionospheric electron density changes on the Doppler flash observed by SuperDARN HF radars. Journal of Geophysical Research: Space Physics, 126, e2021JA029300. doi: 10.1029/2021JA029300
29. Fiori, R.A.D., Chakraborty, S., Nikitina, L. (2022). Data-based optimization of a simple shortwave fadeout absorption model. Journal of Atmospheric and Solar-Terrestrial Physics. doi: 10.1016/j.jastp.2022.105843
30. Chakraborty, S., Baker, J. B. H., Fiori, R. A. D., Ruohoniemi, J. M., and Zawdie, K. A. (2021). A modeling framework for estimating ionospheric HF absorption produced by solar flares. Radio Science, 56, e2021RS007285. doi:10.1029/2021RS007285
31. Chakraborty, S., Ruohoniemi, J. M., Baker, J. B. H., Fiori, R. A. D., Bailey, S. M., and Zawdie, K. A. (2021). Ionospheric Sluggishness: A Characteristic Time-Lag of the Ionospheric Response to Solar Flares. Journal of Geophysical Research: Space Physics, 126, e2020JA028813. doi: 10.1029/2020JA028813
32. Chakraborty, S. and Morley, S. K. (2020). Probabilistic Prediction of Geomagnetic Storms and the Kp Index. Journal of Space Weather and Space Climate. doi:10.1051/swsc/2020037
33. Mukhopadhyay, A., Welling, D. T., Liemohn, M. W., Ridley, A. J., Chakraborty, S., and Anderson, B. J. (2020). Conductance Model for Extreme Events: Impact of auroral conductance on space weather forecasts. Space Weather, 18, e2020SW002551. doi:10.1029/2020SW002551
34. Chakraborty, S., Baker, J. B. H., Ruohoniemi, J. M., Kunduri, B., Nishitani, N., and Shepherd, S. G. (2019). A Study of SuperDARN Response to Co-occurring Space Weather Phenomena. Space Weather, 17. doi:10.1029/2019SW002179

35. Chakraborty, S., Ruohoniemi, J. M., Baker, J. B. H., and Nishitani, N. (2018). Characterization of short-wave fadeout seen in daytime SuperDARN ground scatter observations. Radio Science, 53, 472-484. doi:10.1002/2017RS006488
36. Fiori, R. A. D., Koustov, A. V., Chakraborty, S., Ruohoniemi, J. M., Danskin, D. W., Boteler, D. H., and Shepherd, S. G. (2018). Examining the Potential of the Super Dual Auroral Radar Network for Monitoring the Space Weather Impact of Solar X-Ray Flares. Space Weather, 16. doi:10.1029/2018SW001905

CONFERENCE PAPERS

37. Chakraborty, S., Ruohoniemi, J. M., Baker, J. B. H., Fiori, R., Zawdie, K., Bailey, S., Nishitani, N., and Drob, D. et al. (2020). Sluggishness of the Ionosphere: Characteristic time-lag in Response to Solar Flares, 2020 XXXIIIrd General Assembly and Scientific Symposium of the International Union of Radio Science, Rome, Italy, pp. 1-4, doi: 10.23919/URSIGASS49373.2020.9232206
38. Chakraborty, S., Mukherjee, U. (2015). Circular Micro-strip Antenna Modelling using FDTD method and Design using Genetic Algorithms. CALCON-2015, ID: EC20140700015
39. Chakraborty, S., Mukherjee, U. (2014). Circular Micro-strip (Coax Feed) Antenna Modelling using FDTD Method and Design using Genetic Algorithms: A Comparative Study on Different Types of Design Techniques. ICCCT-2014, IEEE Digital Xplore. doi:10.1109/ICCCT.2014.7001514
40. Chakraborty, S., Mukherjee, U. (2011). Comparative Study of Micro-strip antennas designed by coaxial feed and line feed. ICCCT-2011, IEEE Digital Xplore. doi:10.1109/ICCCT.2011.6075196
41. Chakraborty, S., Mukherjee, U. (2010). Micro-strip antenna optimization using genetic algorithms. ICCCT-2010, IEEE Digital Xplore. doi:10.1109/ICCCT.2011.6075196

INVITED PRESENTATIONS

- Chakraborty, S. (Invited). Characterization of Solar Flare Effects Observed by High Frequency Radar, HAO Colloquium, 12 July, 2023, Online.
- Chakraborty, S. (Invited), Shi, X., Chisham, G., Ruohoniemi, J. M., Baker, J. B. H., Stern, K., and Thomas, E. G. Coordinated Investigation of Antarctic Total Solar Eclipse (TSE) using SuperDARN HF Radars, CEDAR Workshop, Austin, June 2022.
- Maimaiti, M., Chakraborty, S. (Invited). SuperDARN Radars in Space Science Research, CEDAR Workshop, Keystone, June 2017.

SELECTED PRESENTATIONS

- Chakraborty, S., Nishitani, N., Ruohoniemi, J. M., Shi, X., Baker, J. B. H., & Ponomarenko, P. V. (2023, December). Solar flare-induced gradient drift instability observed by SuperDARN HF radars. AGU Fall Meeting Abstracts, 2023, SA31B-2861.
- Shi, X., Chakraborty, S., Baker, J. B. H., Hartinger, M., Lin, D., Wang, W., Sterne, K. T. (2023, December). Quantifying Energy Deposition through Joule Heating from Ionospheric Ultra-Low-Frequency Perturbations using SuperDARN Data. AGU Fall Meeting Abstracts, 2023, SA34A-07.
- Chakraborty, S., Nishitani, N., Ruohoniemi, J. M., Baker, J. B. H., Shi, X., Kunduri, B. Solar Flare effects on High Latitude Electrodynamics, Fall AGU, Chicago, IL, 12-16 December, 2022.
- Boteler, D., Chakraborty, S., Shi, X., Murphy, B., Hartinger, M., Wang, X., Lucas, G., and Baker, J. Modeling Geomagnetic Induction in Submarine Cable, Fall AGU, Chicago, IL, 12-16 December, 2022.
- Chakraborty, S., Nishitani, N., Qian, L., Ruohoniemi, J., Baker, J., McInerney, J. The Role of Flare-Driven Ionospheric Electron Density Changes on the Doppler Flash Observed by SuperDARN HF Radars, STE, Nagoya University, Japan, 27 September, 2022.
- Chakraborty, S., Qian, L., Mabie, J., McInerney, J. M., Mark, S., Erickson, P. J., Earle, G. Origination of Ionospheric G-condition following a Total solar eclipse, TESS, Bellevue, WA, 8-11 August, 2022.
- Chakraborty, S., Nishitani, N., Ruohoniemi, J. M., and Baker, J. B. H. Ionospheric Response to Solar Flares Observed in SuperDARN HF Radars, SoSpIM, Online, 27-28 June, 2022.
- Chakraborty, S., Qian, L., Ruohoniemi, J. M., Baker, J., and McInerney, J. The effects of solar flare-driven ionospheric electron density change on Doppler Flash, SuperDARN Workshop, 2021.
- Chakraborty, S., Baker, J. B. H., Fiori, R. A. D., Ruohoniemi, J. M., & Zawdie, K. A. Testing the role of Dispersion Relation & Collision Frequency Formulations on Estimation of Shortwave-Fadeout (SWF), URSI GASS, Rome, Italy, 2021.

- Chakraborty, S., Baker, J. B. H., Ruohoniemi, J. M., Bailey, S., Fiori, R. A. D., Nishitani, N. A Study of Solar Flare Effects on Mid and High Latitude Radio Wave Propagation using SuperDARN HF Radar, AGU Fall Meeting, Washington DC, 2018.

OUTREACH & ACTIVITIES

- Participated in Summer Camp for pre-college students such as (1) Imagination (rising 7th and 8th graders) and Pathways for Future Engineers (first generation rising 10-12th graders), as a volunteer faculty, organized by the Center for the Enhancement of Engineering Diversity (CEED) at Virginia Tech (Since 2021).
- Participated in Science and Collaborative talks for the freshmen as a volunteer faculty, organized by the Center for the Enhancement of Engineering Diversity (CEED) at Virginia Tech (Since 2021).

INDUSTRIAL WORK EXPERIENCE

A 4.75 years' work experience at Tata Consultancy Services Innovation Labs as IT Analyst. Worked on several projects, based on Home Energy Management System, Device Management System and working different software development modules (PaaS – Platform as a Service) like API and APP gateway, automated user application deployment module, distributed task queue, Sensor data explorer (Cloud-based). Tools and Software used for projects: Python, Java, C (Language), Java-Script, Lua, HTML, XML and JSON, CSS, Scala, Akka [Parallel computational Framework], Redis, and MySQL.

Also contributed significantly to the development of the new Virginia Tech SuperDARN website using Python-flask framework.

REFERENCES

Name	Designation / Institution	Contact Details
Joseph B. H. Baker	Professor, ECE Department Virginia Tech	1185 Perry Street, 453 Whittemore (0111) Virginia Tech, Blacksburg, VA 24061 E: bakerjb@vt.edu T: (540) 231-3355
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